SIALIC-GLYCAN INFLAMMATORY CASCADE · WHITEPAPER

Why Some and Not Others

A three-gate model of individual susceptibility in the Sialic-Glycan Inflammatory Cascade

Martin Hamer, RN (Ret.) · Hamer Glycan Systems · August 2026

Firewall convention. Every substantive claim is labelled ESTABLISHED (published, replicated, peer-reviewed), CONTESTED (published evidence conflicting, disputed, or negative), or HYPOTHESIS (an SGIC proposition not yet tested). No hypothesis is asserted as fact, and no established finding is claimed as original to this framework.

1. The Question

Acne and rosacea together affect on the order of 1.2 billion people. Dietary Neu5Gc, by contrast, is consumed by nearly everyone who eats red meat or dairy, and most humans studied carry circulating antibodies against it. If Neu5Gc exposure were sufficient to produce inflammatory skin disease, the condition would be close to universal. It is not.

Any account of the Sialic-Glycan Inflammatory Cascade must therefore explain partial penetrance: why a common exposure, acting on a near-universal antibody response, produces symptomatic disease in a substantial minority and nothing in the majority. This paper proposes that susceptibility requires three independent conditions to coincide, and that failing any one of them is sufficient for protection.

Plain-language summary. Three gates must line up. First, your particular antibodies must recognise the specific bovine form of the sugar — the framework proposes that most people′s do not. Second, something must breach the gut barrier at the right moment, so the sugar arrives alongside bacterial fragments that put the immune system on alert. Third, the local brake on inflammation in the oil glands of the skin must be weak enough that an ordinary immune nudge is not switched off. Fail any one gate and there is no disease. Pass all three and there is.

2. Gate One — Antibody Repertoire

2.1 ESTABLISHED — Anti-Neu5Gc antibodies are diverse and vary between individuals

Neu5Gc cannot by itself fill the binding site of an antibody. The paratope accommodates several linked monosaccharides, so what an antibody recognises is an extended epitope comprising Neu5Gc together with the underlying glycan chain. Structural diversity arises from 9-O-acetylation, differences in sialyl linkage, and the complexity of the glycan beneath, producing many distinct Neu5Gc-containing epitopes.

Padler-Karavani and colleagues, working with Nissi and Ajit Varki, used natural and chemoenzymatically synthesised glycans to show that normal humans carry an abundant and diverse

spectrum of anti-Neu5Gc antibodies directed against a variety of Neu5Gc-containing epitopes, most commonly of the IgG class. Subsequent work established that individual antibody profiles differ between people, and that conventional ELISA assays directed against Neu5Gc alone, or against a single Neu5Gc-glycan, may be inadequate to define risk. A 2024 systematic review noted that this presentation on diverse underlying scaffolds distinguishes anti-Neu5Gc antibodies from anti-alpha-Gal antibodies, which target a single epitope and are reliably found across the entire human population — offering a probable explanation for why the anti-Neu5Gc response remains contested.

The consequence for this framework is direct. Neu5Gc content is the wrong measurement. Two foods with identical Neu5Gc concentrations may present entirely different epitopes, and two individuals with identical dietary exposure may carry antibody repertoires that recognise entirely different subsets of those epitopes. Severity should track epitope match, not antigen quantity.

2.2 Supporting observation — four decades of species-specific negative controls

The author has recorded consistent inflammatory response to bovine dairy and beef across four decades, with consistent absence of response to goat, sheep, pork and fish — all of which contain Neu5Gc. Goat milk in particular carries a higher proportion of its sialic acid as Neu5Gc than bovine milk does, and has never produced a recorded reaction.

Under a content model this is inexplicable. Under an epitope model it is the expected result: non-bovine mammals synthesise Neu5Gc using their own sialyltransferase repertoires and install it on their own glycan scaffolds. The monosaccharide is identical; the epitope is not.

2.3 HYPOTHESIS — Secondary processing is re-presentation, not detoxification

Earlier SGIC drafts proposed that Neu5Gc passing through a second mammal is rendered non-inflammatory by that animal's metabolism. The observation is sound but the mechanism was misstated. Feed regulations under 21 CFR 589.2000 permit bovine milk and milk proteins, blood and blood products, gelatin, and tallow containing no more than 0.15 percent insoluble impurities in ruminant feed, and ruminant is defined to include sheep and goats. Kid goats are commonly reared on bovine-whey-based milk replacer. Bovine Neu5Gc exposure in goats is therefore real and substantial.

Nothing detoxifies the sugar. Ingested bovine glycoconjugates are catabolised to free Neu5Gc, which enters the recipient animal's own CMP-sialic acid pool and is reinstalled by that animal's own sialyltransferases onto its own scaffolds. The molecule survives unchanged; the epitope does not. Secondary processing re-presents Neu5Gc rather than destroying it.

A second filter applies to ruminants specifically. Feed passes through a fermentation chamber rich in bacterial sialidases before reaching the abomasum, so bovine glycoconjugates are likely degraded substantially before absorption. This filter does not apply to pigs or fish, which rely on re-presentation alone.

2.4 CONTESTED — The blood group hypothesis

It has been proposed within this framework that ABO or Rh type governs susceptibility. The published evidence does not support it, and runs in the opposite direction.

A Turkish retrospective study compared 498 clinically confirmed acne patients against 419 blood donors from the same geographic area. Blood group AB was significantly over-represented among patients, while blood group O was significantly lower in the patient group than in controls. No statistically significant difference was found in Rh distribution. A later study of 380 acne patients against 1,000 controls found a correlation between acne severity and ABO group but again none with Rh factor, with type A associated with the higher rate of severe disease. Earlier reports have been contradictory.

The arithmetic is also unfavourable. O-negative individuals constitute approximately five to six percent of the global population, and considerably less in East Asia — on the order of 450 million people. Even universal penetrance among them would not account for 1.2 billion cases.

> **Position taken.** The blood group hypothesis is not supported and should be set aside. It is retained in this document rather than deleted because the question it was raised to answer — individual variation in susceptibility — is the right question, and Section 2.1 answers it better. A registry field capturing ABO and Rh remains worth adding, at negligible cost, so that the matter can be settled from the author's own data rather than by argument.

3. Gate Two — The Priming Event

3.1 HYPOTHESIS — Barrier permeability as adjuvant

Antigen alone is not sufficient to convert quiet immune clearance into symptomatic inflammation. The proposition here is that a secondary priming event is required, and that increased intestinal permeability supplies it.

Where the barrier is compromised — by wheat amylase-trypsin inhibitors, dysbiosis, stress, or gliadin-mediated zonulin release — intact Neu5Gc-bearing glycoproteins enter circulation concurrently with bacterial antigens. The bacterial material provides an adjuvant effect, shifting the immune response from tolerant clearance to active inflammatory engagement. Neu5Gc supplies the target; the barrier breach supplies the alarm.

This is the same two-signal architecture set out in the companion whitepaper in this series, Visceral Adiposity, Wheat Amylase-Trypsin Inhibitors and Dietary Neu5Gc: ATIs prime TLR4 on myeloid cells, and anti-Neu5Gc IgG bound to incorporated antigen engages Fc-gamma receptors and complement on those same primed cells. TLR priming followed by Fc crosslinking is a well-characterised superadditive combination. The claim advanced here is not a new mechanism but a new target tissue.

The clinical corollary is that exposure and symptom need not be temporally tight. A person may consume bovine dairy for months without incident and then react sharply during a period of gastrointestinal disturbance, illness, or stress. Under a simple dose model this looks like inconsistency. Under a two-signal model it is the expected pattern.

4. Gate Three — The Local Brake

4.1 ESTABLISHED — Complement regulatory proteins and how they act

When antibody binds to a cell surface, the complement cascade is initiated. Human cells restrain it through membrane-bound regulatory proteins. CD55, or decay-accelerating factor, dissociates the C3 and C5 convertases. CD59 acts later, binding the C5b-8 and C5b-9 intermediates and preventing full assembly of the membrane attack complex. These operate at two distinct points in the cascade and are not interchangeable.

An important refinement. At sublytic levels the membrane attack complex does not destroy the cell. It signals — activating the NLRP3 inflammasome and driving IL-1beta release. This matters to the present model because it produces inflammation without requiring tissue destruction, and because IL-1beta is the same cytokine implicated in the ATI arm of this framework. Two entry points, one output.

4.2 HYPOTHESIS — Locally reduced regulator expression confers susceptibility

The proposition is that individuals with lower baseline expression of CD55 and CD59 on endothelial and sebaceous cells cannot terminate complement activation triggered by anti-Neu5Gc antibody binding, and sustain localised inflammation where others do not.

The susceptibility must be local rather than systemic. Globally deficient complement regulation produces haemolysis, as in paroxysmal nocturnal haemoglobinuria, not acne. Any version of this hypothesis that implies systemic deficiency is refuted by the absence of systemic disease in the affected population.

4.3 Precedent — bullous pemphigoid

A closely analogous mechanism has been demonstrated in another antibody-mediated skin disease. CD55 was found to be significantly lower in the lesional skin of bullous pemphigoid patients than in healthy control skin, while showing no difference in the peripheral blood cells of the same patients. Depletion of CD55 in cultured human keratinocytes enhanced autoantibody-mediated complement activation, and exogenous recombinant CD55 blocked that activation in both skin sections and keratinocytes. TNF-alpha and IFN-gamma, both elevated in patient sera, were shown to downregulate CD55 via ERK1/2 signalling.

This precedent supplies three things: proof that locally reduced complement regulation can determine susceptibility in skin, a demonstration that the deficit can be invisible in blood, and a study design that can be copied directly.

> **Caution to state before a reviewer does.** Because TNF-alpha downregulates CD55, reduced regulator expression in lesional skin may be a consequence of inflammation rather than its cause. The bullous pemphigoid literature carries the same ambiguity. This does not greatly damage the model, since either direction of causation yields a self-reinforcing loop — which is the architecture the SGIC framework proposes in any case — but the ambiguity must be acknowledged rather than resolved by assertion.

5. Why the Pilosebaceous Unit

A three-gate model must still explain tissue selection. Why does this cascade manifest in the oil glands of the face rather than uniformly across the body?

5.1 HYPOTHESIS — Membrane turnover concentrates Neu5Gc in sebocytes

Sebocytes are holocrine. The entire cell disintegrates in order to release sebum. This entails membrane synthesis and turnover on a scale matched by few tissues in the body, and membrane synthesis is precisely where dietary Neu5Gc is incorporated — activated to CMP-Neu5Gc and installed by sialyltransferases onto nascent glycoproteins and glycolipids in the Golgi. The pilosebaceous unit would therefore be among the most Neu5Gc-loaded structures available, not by any special affinity but by sheer rate of membrane production.

This also accounts for the timing. Sebaceous activity is androgen-driven and rises sharply at puberty, which is when the substrate for this cascade first becomes abundant. It further predicts that the framework should apply most strongly to sebaceous-rich sites — face, chest, upper back — which is the observed distribution of both acne and rosacea.

6. The Composite Model

Gate	Requirement for disease	Status	Test
One — repertoire	Antibody spectrum includes bovine-type Neu5Gc epitopes	Established that repertoires vary; bovine specificity in this subject is hypothesis	Sialoglycan microarray on subject serum
Two — priming	Barrier permeability delivers antigen with bacterial adjuvant	Hypothesis; components established separately	Symptom correlation with permeability markers
Three — brake	Locally reduced CD55/CD59 in pilosebaceous tissue	Hypothesis; precedent established in bullous pemphigoid	CD55/CD59 IHC, lesional vs non-lesional skin
Target	High membrane turnover in holocrine sebocytes	Established sebaceous physiology; Neu5Gc loading is hypothesis	Neu5Gc IHC on sebaceous tissue

The model is multiplicative rather than additive. Protection at any single gate is sufficient, which is what a partial-penetrance account requires. It also predicts that interventions need not address all three: reducing exposure closes gate one from the supply side, and restoring barrier integrity closes gate two, without any need to modify complement regulation.

7. Predictions and Falsification

- P1 — Subject serum tested on a sialoglycan microarray will bind bovine-type Neu5Gc glycotopes preferentially over caprine and porcine ones. This is the crucial experiment: a negative result indicates the operative antigen is not Neu5Gc but a bovine protein, and the framework requires rebuilding.
- P2 — CD55 and CD59 expression will be lower in lesional than non-lesional pilosebaceous tissue in acne patients, and lower than in matched controls, while peripheral blood expression is unchanged.
- P3 — Symptom onset will correlate with markers of intestinal permeability rather than with dietary exposure timing alone.
- P4 — Neu5Gc will be detectable in sebaceous tissue at concentrations exceeding those in adjacent dermis, reflecting differential membrane turnover.
- P5 — ABO and Rh distribution in the SGIC registry cohort will not differ from population baseline. A positive finding here would contradict Section 2.4 and require its revision.

8. Objections Not Resolved

- Acne is near-universal in adolescence. The relevant question may be severity rather than presence, and this model is better suited to explaining severe or persistent disease than any occurrence of a comedone.
- Androgens, heritability, and Cutibacterium acnes phylotype are established contributors with substantial explanatory power. This model must earn its place alongside them rather than displacing them, and no claim is made here that it is the dominant factor.
- Three hypothesised gates, each individually untested, multiply uncertainty rather than dividing it. A model with three free parameters can accommodate almost any observation, which is a liability and not a strength. The predictions in Section 7 exist to constrain it.
- The subject-as-investigator problem applies throughout Section 2.2. Four decades of self-observation are the origin of this framework and cannot also serve as its confirmation.

Selected References

Padler-Karavani V, Yu H, Cao H, Chokhawala H, Karp F, Varki N, Chen X, Varki A. Diversity in specificity, abundance, and composition of anti-Neu5Gc antibodies in normal humans: potential implications for disease. Glycobiology. 2008;18(10):818-830.

Padler-Karavani V, Hurtado-Ziola N, Pu M, Yu H, Huang S, et al. Human xeno-autoantibodies against a non-human sialic acid serve as novel serum biomarkers and immunotherapeutics in cancer. Cancer Res. 2011;71(9):3352-3363. doi:10.1158/0008-5472.CAN-10-4102

Hutton E, Scott E, Robson C, Signoret N, Fascione MA. A systematic review reveals conflicting evidence for the prevalence of antibodies against the sialic acid xenoautoantigen Neu5Gc in humans and the need for a standardised approach to quantification. Front Mol Biosci. 2024;11:1390711. doi:10.3389/fmolb.2024.1390711

Qiao P, Dang EL, Fang H, Zhang JY, Li B, Shen SX, Luo YX, Lei J, Shao S, Qiao HJ, Wang G. Decreased expression levels of complement regulator CD55 contribute to the development of bullous pemphigoid. Oncotarget. 2018;9(85):35517-35527. doi:10.18632/oncotarget.21216

Bas Y, et al. The relationship between ABO blood groups and acne vulgaris. Saudi J Med Med Sci. 2016;4(1). PMID 30787691.

Gurok NG. The correlation between ABO blood types and acne vulgaris severity. J Cosmet Dermatol. 2023. PMID 36866960.

Zevallos VF, Raker V, Tenzer S, et al. Nutritional wheat amylase-trypsin inhibitors promote intestinal inflammation via activation of myeloid cells. Gastroenterology. 2017;152(5):1100-1113.e12.

Tangvoranuntakul P, Gagneux P, Diaz S, et al. Human uptake and incorporation of an immunogenic nonhuman dietary sialic acid. PNAS. 2003;100(21):12045-12050.

21 CFR 589.2000 — Animal proteins prohibited in ruminant feed.